DATA_MODEL

Status: Draft v1. Database: MongoDB. Files: local disk.

1. Design rules

- Every user-owned document carries userId. Every query filters by userId.
- Workflow state changes only through conditional updates (filter on current state and version).
- Large files (snapshots, V2, diffs) live temporarily on server disk. The permanent work lives on the GitHub work branch. MongoDB stores only metadata and paths.
- No secrets in logs or audit entries.
- Single-document atomic updates are preferred. Multi-document transactions need a replica set, so avoid them unless a case needs them.

2. Collections

users

Field	Notes
_id	ObjectId
githubId	Unique, from GitHub
githubLogin	Username at last login
avatarUrl	Optional
encryptedToken	GitHub token, encrypted at rest. Never returned by API.
tokenScopes	Scopes granted
status	active or banned
createdAt, lastLoginAt	Dates
deletionRequestedAt	Set when user asks for data delete

Indexes: unique githubId.

sessions

Session id (random), userId, createdAt, expiresAt, userAgent hash. TTL index on expiresAt. Cookie is httpOnly, secure, sameSite.

workflows

Field	Notes
_id, userId	Owner
repo	owner, name, githubRepoId, defaultBranch, isPrivate (must be false), ownerType (must be User)
sourceSha	Exact commit pinned at F4
workBranch	Work branch created at F4 on GitHub: obsidian/enhance-REPO, suffix -2, -3 if the name exists
workBranchHeadSha	Latest commit SHA on the work branch, updated after each Obsidian commit
state	See WORKFLOW_STATE_MACHINE
version	Integer, increases on every update
snapshot	path, sizeBytes, fileCount, createdAt
profile	languages, frameworks, packageManagers, evidence list
v2	path, createdAt, manifestId
failure	code, message, at
prPermission	grantedAt, grantedBy, sourceSha, planId
pr	number, url, createdAt
createdAt, updatedAt, expiresAt	expiresAt drives retention cleanup

Indexes: userId plus createdAt, userId plus state, unique active workflow per (userId, repo.githubRepoId) so two workflows cannot run on one repo at once (partial index on non-terminal states).

findings

workflowId, userId, type (bug, security, enhancement, preexisting), severity, title, explanation, evidence (filePath, lineStart, lineEnd, excerpt), confidence (verified or uncertain), source (rule or ai), createdAt. Findings are read-only reports and are never linked to automatic fixes.

Indexes: workflowId, userId.

plans

workflowId, userId, sourceSha, status (draft, approved, superseded), operations array, createdAt.

Each operation: opId, type (edit, add, delete, rename), category (ui, animation, performance, security, structure, other), files (relative paths), risk (low, medium, high), expectedChange, validationNotes, selected (boolean), requiresExtraConfirmation (true for high risk or delete).

Operations come from AI suggestions but are stored only after schema and scope validation.

approvals

_id, userId, workflowId, planId, sourceSha, approvedOpIds (list), extraConfirmedOpIds (list), planHash (hash of approved operation content), createdAt, status (valid, stale, consumed). Immutable once written except status. Unique index on (workflowId, planHash) for idempotency.

manifests

workflowId, userId, sourceSha, approvalId, entries (path, change type, beforeHash, afterHash, opId), createdAt. Every changed file must map to an approved opId.

staticChecks

workflowId, userId, results (per file: parse ok or error, message), diffMatchesManifest (bool), unapprovedChanges (list), createdAt.

auditLogs

id, userId, workflowId, action (login, repo_selected, approved, executed, push_permitted, pushed, banned, data_deleted, etc.), result, correlationId, at. Append-only by convention (the application never updates or deletes entries except by retention policy). No secrets, no code content.

Indexes: userId plus at, workflowId.

usageCounters

userId, day, workflowsStarted, aiTokensUsed, aiCostEstimate. Used for rate limits and AI cost limit.

jobs (metadata)

BullMQ stores jobs in Redis. MongoDB keeps jobId, workflowId, step, attempt and status so state can be rebuilt if Redis is lost.

3. Disk layout

backend/storage/USERID/WORKFLOWID/ contains snapshot/, v2/, diff.patch, manifest.json. This folder is gitignored and is only a temporary working copy. It is deleted when the job ends or fails. The permanent copy is the commits on the GitHub work branch. Paths are generated by the server only. User input is never used as a path.

4. Consistency requirements

Case	Rule
State change	updateOne where _id, userId, state equals expected and version equals expected
Approval	Insert with unique key. Repeat click returns the same approval.
Execute	Job id derived from workflowId plus approvalId, so duplicates do not run twice.

Case	Rule
Push	Allowed only from PUSH_PERMITTED and only once. Opening the PR is idempotent by the pr field. Commits are tracked by commit SHA so a retry does not duplicate them.
Stale	If remote SHA differs from sourceSha, set STALE and mark approvals stale.

5. Retention and deletion

- Snapshots and V2 are deleted after a retention period (number to be decided) or when the user deletes the workflow. The GitHub work branch belongs to the user and is not deleted automatically.
- Data delete removes workflow records, findings, plans, approvals, manifests, checks and disk files, then verifies the folder is gone before marking done.
- Audit entries keep only non-sensitive metadata. Policy for deleted users: keep minimal anonymous metadata or remove (open decision).
- Backups of server work files are not needed because the work branch on GitHub is the permanent copy.

6. Open decisions

Retention period length. Audit handling after user deletion. Exact limits (repo size, file count). Whether a replica set is needed later.